B.TECH MID-TERM EVALUATION · DEPARTMENT OF DATA SCIENCE · IIT PALAKKAD

A probabilistic model of motor subtype progression in Parkinson's disease

The clinical rule tells us a patient's subtype **today**. We ask a different question: when the recorded subtype changes, how much of that is the disease progressing — and how much is the measurement?

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Parkinson's Progression Markers Initiative (PPMI)

One number, cut into three

In the MDS-UPDRS examination, eleven items measure tremor and five measure gait and postural difficulty. The Stebbins (2013) rule compares their averages, and two fixed thresholds turn that comparison into a label. We use this definition unchanged.

R = mean of 11 tremor items ÷ mean of 5 gait / postural items

R ≥ 1.15 → TD · 0.90 < R < 1.15 → Indeterminate · R ≤ 0.90 → PIGD



Drawn to scale. The **Indeterminate** band is only **0.25 units wide** on a quantity that runs from 0 to 11.36 — this turns out to matter a great deal.

What the rule answers

"What subtype does this patient appear to have today?"

What it cannot answer

"Has this patient's subtype actually changed — or did the examination just come out differently?"

Stebbins et al., Movement Disorders 2013

The dataset we built

Recomputed from the 16 raw MDS-UPDRS items rather than taken off the shelf — sporadic PD only, OFF-medication exams — then checked back against PPMI's own labels.

1,080
sporadic PD patients

5,774
patient visits

4,441
visits with a subtype label

934
patients with ≥2 labelled visits — the analysis cohort

LABELS VALIDATED AGAINST PPMI'S OWN

Reference	Classes	Agreement
Curated cut `td_pigd`	2	4,431 / 4,431 = 100%
Archived `td_pigd_old`	3	1,595 / 1,597 = 99.87%

A TRAP THAT HAD TO BE FOUND FIRST

"Unable to Rate" is coded 101, in the same field as a real 0–4 score. Averaged as a score, one such entry lifts the five-item PIGD mean to roughly 20, drives the ratio to zero, and relabels a tremor-dominant visit as PIGD. Postural stability alone carries **3,206** of them.

What the records actually show

3,361 consecutive labelled pairs from 934 patients, median interval 1.0 year. Rows are the current visit, columns the next.

Subtype at this visit	→ TD	→ Indeterminate	→ PIGD
TD (n = 2,261)	0.808	0.080	0.112
Indeterminate (n = 334)	**0.440**	**0.192**	**0.368**
PIGD (n = 766)	0.184	0.114	0.702

Indeterminate barely persists

Only **19.2%** of Indeterminate visits are followed by another. It behaves like a crossing rather than a category a patient belongs to.

PIGD is not a destination

29.8% of PIGD visits are followed by a non-PIGD assessment, **18.4%** straight back to TD. Treating TD → PIGD as one-way is contradicted by the data.

But the drift is real

253 changes run TD → PIGD against **141** the other way ($p = 1.8 \times 10^{-8}$). Something genuine is moving underneath.

A single recorded change is weak evidence

35.2%

of all observed subtype changes **reverse at the very next visit**. For changes into the Indeterminate band, 50–56%.

51.7%

of apparent PIGD conversions disappear if we simply require the new subtype to hold for two consecutive visits — **263 drops to 127**.

73%

of consecutive visit pairs move a patient's own ratio by **more than the entire width** of the Indeterminate band.

Within a single patient the ratio has a standard deviation of **1.22** against a band **0.25** wide, and a median visit-to-visit move of **0.68**.

So a patient sitting near a threshold will cross it from time to time **for reasons unrelated to progression**. Band membership was always going to be noise-dominated.

The label is a noisy reading. Counting recorded changes therefore over-counts progression — which is what pushed us towards a model that treats the label as a measurement rather than as the truth.

We tested the simple model first — and it failed

A Markov chain on the observed label assumes the current label is all you need: where the patient came from should not matter. Take only the visits currently labelled **TD**, and ask how likely PIGD is next.

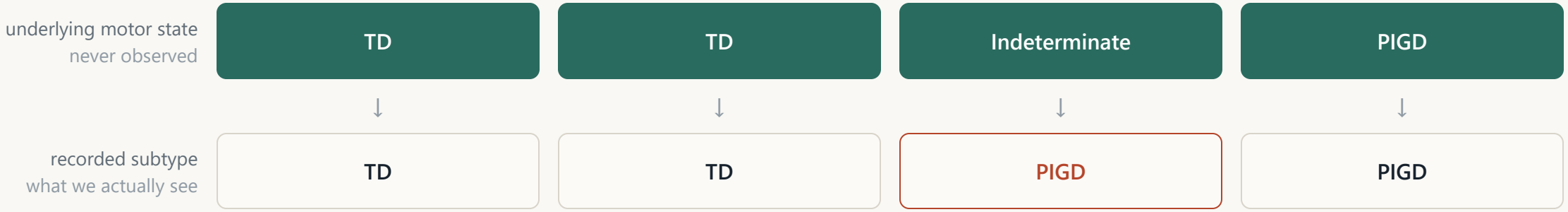


Same label today, four and a half times the risk tomorrow. χ^2 = **156.6** on 4 df — and the test rejects independence for **all three** current labels, not just TD (Indeterminate p = 0.004, PIGD p < 10^{-5}).

The current label is **not a sufficient summary of the patient's past**. That is exactly the signature of an underlying state being read through a noisy measurement — which is what motivated the hidden Markov model.

χ^2 test on labelled triples, training patients only

Separating the state from the label



Two things estimated at once

How fast the underlying state changes, **and** how reliably each state gets recorded. The top row moves smoothly; the bottom row — all we see — appears to jump.

Time enters directly

Visits are irregular, so the latent chain runs in **continuous time** and the elapsed gap is an input. Compared against a discrete chain, an interval-aware chain, and a null model.

One input we choose

A minimum recording reliability (**0.5**) is imposed, or the fit degenerates. Conclusions hold across a range of values, but it is an assumption, not an estimate.

What the model estimates once noise is accounted for

FINDING 1

Progression is sequential

TD → Indeterminate runs at **0.094** per year; TD → PIGD directly at **0.006** — sixteen times rarer. PIGD → TD directly is zero. The model recovered an ordering it was never given.

FINDING 2

One label is far less reliable

Underlying state	Recorded right
TD	94.1%
PIGD	84.2%
Indeterminate	**50.0%**

FINDING 3

States outlast their labels

State	Label	Underlying
TD	3.4 y	10.1 y
Indet.	0.4 y	2.1 y
PIGD	2.3 y	16.5 y

> The underlying state appears **three to seven times more persistent** than the recorded subtype suggests — so on this model's reading, most of the apparent instability is variability in the examination rather than change in the patient. These are model estimates, not measurements.

Validation: leave one patient out

Each of the 934 patients is withheld in turn; the model is refitted from scratch on the other 933 and then predicts that patient's visits one at a time, using only the history available before each prediction. Nothing about the held-out patient touches the fit.

75.4%

2,534 of **3,361** predictions correct
95% CI (patient bootstrap) **73.5 – 77.3%**

Model, identical predictions	Accuracy	Log-lik / pred	Brier
Hidden Markov model	**75.4%**	**−0.673**	**0.369**
Markov chain on observed labels	74.7%	−0.707	0.390
Interval-aware (continuous-time) chain	74.5%	−0.711	0.393
Persistence — "stay where you are"	72.3%	—	0.555
Null — ignores the current state	62.9%	−0.876	0.521

What this measures is agreement with **the next label our own classification rule produces** — not an independently verified diagnosis. It is not clinical validation, and the gain over simply saying "nothing changes" is real but modest: **3.1 points**.

Patient-level splitting is essential: visits from one patient are correlated, so splitting visits rather than patients would leak a patient's own future into their training data. **All ten leakage and consistency checks passed.**

Every fold refits from the same seed; the run is fully reproducible

Is 75.4% robust? Two stress tests

Having a number, we tried to break it. We changed the thing being predicted, and we changed the size of the model. Both came back negative — and both were informative.

TEST 1 · NARROW THE INDETERMINATE BAND TO 0.91–1.13

On the same 3,361 predictions	Original band	Narrow band
Visits labelled Indeterminate	441	143
HMM accuracy	75.4%	78.0%
Persistence baseline	72.3%	76.9%
HMM − persistence	**+3.1**	**+1.2**

Accuracy went **up** by 2.6 points while the model's advantage over "nothing changes" was **cut in half**. Shrinking the hardest class simply removes the cases everything got wrong.

TEST 2 · FIVE LATENT STATES INSTEAD OF THREE

More capacity did not help

Accuracy **75.1%** — no better than three states, on identical predictions.

Only **404 of 934** folds converged (43%).

Six deliberately dispersed starting points produced **six different solutions**. The extra states are not identified by the data.

A higher accuracy number is not automatically a better model. Both tests moved the number without improving the science — which is why we quote the **margin over the baseline** rather than accuracy on its own.

A second view: the continuous ratio

The proposal also specified a hierarchical Bayesian growth model on the ratio itself. We implemented it as written — not to replace the HMM, but because it answers a question the HMM cannot.

What only this model gives

A progression rate on the clinical scale, with uncertainty: the population ratio declines **0.093 per year** (95% CI 0.068–0.116). From a mean baseline of 2.29 the average patient reaches the TD threshold after about **12.3 years**. It also gives each patient their own slope.

Where it breaks

Its residual SD is **1.138 — 4.6× the entire Indeterminate band**, so band membership was never recoverable from it. It loses to the HMM on next-visit classification in **6 of 6** splits, and a Normal distribution is wrong for a quantity that is non-negative and exactly zero at 7% of visits.

Per-patient reliability is 0.805 for where a patient starts, but only **0.050 for their slope** — with a median of three visits per patient, an individual's rate of progression is essentially unrecoverable. We keep both models because they answer different questions.

Our current reading: **the noise and the signal coexist at comparable size.** Not that subtype progression is an artefact, and not that every recorded change is real — but that a single recorded change cannot be read as either.

Where this goes next

Four directions, coming from the limitations above and from our discussion with the domain researcher.

1	**Model the two subscores separately, not their ratio**	The ratio compresses two quantities into one and loses severity: **2/4 and 10/20 are both 0.5**, but they are not the same patient. Modelling tremor and gait jointly should retain that, and it also recovers the **725 visits** where the ratio is undefined because the PIGD score is zero.
2	**Let progression rate vary with time**	Every model here assumes the rate is constant over follow-up. We want to compare early, middle and late follow-up directly. **This is a hypothesis to test** — we have not established it.
3	**Cluster longitudinal patterns**	If per-patient trajectory information can be extracted reliably, do patients fall into distinct progression patterns that might correspond to meaningful phenotypes? Not "run clustering" — the question is whether such groups exist at all.
4	**Why and when does the HMM fail?**	Vary initialisation, restarts, the emission floor and the number of latent states, and compare fitted rates, emission matrices, likelihoods and held-out predictions. The five-state result on the previous slide is the reason this matters: we need to know **when** a fit is trustworthy, not only whether it converged.

IN ONE SENTENCE

Motor subtype progression is a gradual underlying process, observed through an imperfect clinical measurement.

A single observed subtype change is weak evidence that a patient has changed. A sustained one is considerably stronger. Where a result depends on counting subtype changes, it is worth checking how sensitive it is to that distinction — including our own.

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Data from PPMI